



Sri Chaitanya IIT Academy, India.

A.P. T.S. KARNATAKA TAMILNADU MAHARASTRA DELHI RANCHI

A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

SEC: Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT

JEE-MAIN

Date: 04-01-2026

Time: 09.00Am to 12.00Pm

GTM-09

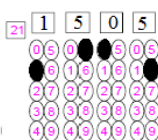
Max. Marks: 300

IMPORTANT INSTRUCTION:

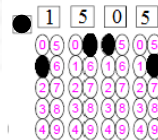
1. Immediately fill in the Admission number on this page of the Test Booklet with **Blue/Black Ball Point Pen** only.
 2. The candidates should not write their Admission Number anywhere (except in the specified space) on the Test Booklet/ Answer Sheet.
 3. The test is of **3 hours** duration.
 4. The Test Booklet consists of **75 Questions**. The maximum marks are **300**.
 5. There are **three** parts in the question paper 1,2,3 consisting of **Mathematics, Physics and Chemistry** having **25 Questions** in each subject and subject having **two sections**.
(I) Section-I contains **20 Multiple Choice Questions** with only one correct option.
Marking scheme: +4 for correct answer, **0** if not attempt and **-1** in all other cases.
(II) Section-II contains **05 Numerical Value Type Questions**.
- The Answer should be within **0 to 9999**. If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i.e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11**).

To cancel any attempted question bubble on the question number box.

For example: To cancel attempted Question 21. Bubble on 21 as shown below



Question Answered for Marking



Question Cancelled for Marking

Marking scheme: +4 for correct answer, **0** if **not attempt** and **-1** in all other cases.

6. Use **Blue / Black Point Pen only** for writing particulars / marking responses on the Answer Sheet. **Use of pencil is strictly prohibited.**
7. No candidate is allowed to carry any textual material, printed or written, bits of papers, mobile phone any electron device etc, except the Identity Card inside the examination hall.
8. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
9. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
10. **Do not fold or make any stray marks on the Answer Sheet**

Name of the Candidate (in Capital): _____

Admission Number:

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Candidate's Signature: _____

Invigilator's Signature: _____

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Page 1



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**THE PERFECT HAT-TRICK WITH ALL-INDIA RANK 1 IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023**

JEE MAIN 2023

SINGARAJU VIKRAM KORDINIA

RANK 1

341

RANK

1

JEE Advanced 2023

VAVILALA CHIVILS REDDY

RANK 1

341

RANK

1

NEET 2023

BORA VARUN

RANK 1

726

RANK

1

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MATHEMATICS : Total Syllabus

PHYSICS : Total Syllabus

CHEMISTRY : Total Syllabus

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Page 2



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**THE PERFECT HAT-TRICK WITH ALL-INDIA RANK 1
IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023**

**JEE MAIN
2023**

**SINGARAJU
VENKAT KOURDINYA**

ALL INDIA RANK 1

SCORE 300

RANK

1

**JEE Advanced
2023**

**VAVILALA
CHIVILAS REDDY**

ALL INDIA RANK 1

SCORE 341

RANK

1

**NEET
2023**

**BORA VARUN
CHAKRAVARTI**

ALL INDIA RANK 1

SCORE 726

RANK

1

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**MATHEMATICS****Max Marks: 100****SECTION-I
(SINGLE CORRECT ANSWER TYPE)**

This section contains **20 Multiple Choice Questions**. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

- In triangle $\triangle ABC$, if $\vec{AB} = \frac{\vec{u}}{|\vec{u}|} - \frac{\vec{v}}{|\vec{v}|}$ and $\vec{AC} = \frac{2\vec{u}}{|\vec{u}|}$ where $|\vec{u}| \neq |\vec{v}|$, then $\sum \cos(2A)$ equals
 1) -2 2) -3 3) -1 4) -4
- The perpendicular distance of point (2,0,-3) from line which passes through the point (0,2,-4) and perpendicular to the lines $\vec{r} = (-3\hat{i} + 2\hat{k}) + \lambda(2\hat{i} + 3\hat{j} + 5\hat{k})$ and $\vec{r} = (\hat{i} - 2\hat{j} + \hat{k}) + \mu(-\hat{i} + 3\hat{j} + 2\hat{k})$, where $\lambda, \mu \in R$ is
 1) $\frac{\sqrt{219}}{3}$ 2) $\frac{\sqrt{78}}{3}$ 3) $\frac{\sqrt{52}}{3}$ 4) $\frac{\sqrt{126}}{3}$
- Let $A_1, A_2, \dots, A_n$ be a regular n - sided polygon inscribed in a circle. Three points A_i, A_j, A_k are randomly chosen, where i,j,k are distinct integers between 1 and n, both inclusive. Denote by P_n the probability that the triangle $A_i A_j A_k$ is obtuse. Which of the following statements is/are incorrect?
 1) P_5 equals $\frac{1}{2}$ 2) P_{10} equals $\frac{1}{2}$ 3) P_{10} equals $\frac{2}{3}$ 4) P_{15} equals $\frac{9}{13}$
- Let $P\left(\frac{2\sqrt{3}}{\sqrt{7}}, \frac{6}{\sqrt{7}}\right)$, Q,R and S be four points on ellipse $9x^2 + 4y^2 = 36$. Let PQ & RS be mutually perpendicular chords and these pass through the centre of ellipse then the value of $\left[\frac{50}{PQ^2} + \frac{50}{RS^2}\right] = \text{---}$ where $[\cdot]$ denotes GIF
 1) 4 2) 5 3) 6 4) 7
- $\int \frac{x(x \tan^{-1} x + (\ln x)(\ln(\ln x))) + \tan^{-1} x}{(x^3 + x) \ln x} dx = f(x) + c$ where 'c' is integration constant & $f(e) = 0$
 Then $\left[\lim_{x \rightarrow 1^+} \frac{f(x)}{\tan\left(\frac{\pi x}{2}\right)} + \frac{11}{10} \right] = \text{---}$, where $[\cdot]$ denotes GIF
 1) 2 2) 1 3) 0 4) -1

**JEE MAIN****2023**SINGARAJU
VENKAT KORDINYA
AIR 1000000
RANK 1000000
JEE MAIN 2023**RANK****1****JEE Advanced****2023**VAVILALA
CHIVILAS REDDY
AIR 1000000
RANK 1000000
JEE ADVANCED 2023**RANK****1****NEET****2023**BORA VARUN
CHAKRAVARTI
AIR 1000000
RANK 1000000
NEET 2023**RANK****1**



6. Let $y = f(x)$ be a differentiable function satisfying $\int_2^x f(t)dt + 2 = \frac{x^2}{2} + \int_x^2 t^2 f(t)dt$ then
- $$\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{f(x) + x^9 - x^3 + x + 1}{\cos^2 x} dx =$$
- 1) 2 2) 3 3) 4 4) 0
7. Let 'q' be the maximum integral value of p in $[0, 10]$ for which the roots of the equation $x^2 + px + \frac{5p}{4} = 0$ are rational then the area of region $\{(x, y) : 0 \leq y \leq (x - q)^2, 0 \leq x \leq q\}$ is (in square units)
- 1) 243 2) 25 3) $\frac{125}{3}$ 4) 164
8. A classroom has a row of 35 coat hooks. Paulina likes coats to be equally spaced, so that there is the same number of empty hooks before the first coat, after the last coat, and between every coat and the next one. Suppose there is at least 1 coat and at least 1 empty hook. How many different numbers of coats can satisfy Paulina's pattern?
- 1) 2 2) 4 3) 5 4) 7
9. The value of S, where $S = \sum_{r=0}^{100} {}^{100}C_r (-1)^r (2026)^r$, satisfies
- 1) $S < 1$ 2) $1 < S < 100$ 3) $S > 1000$ 4) $S > 2026$
10. The number of real solutions of the equation $x^3 - 6x^2 + 11x - 6 = 0$ is
- 1) 1 2) 2 3) 3 4) 0
11. Let $p = \tan\left(\frac{5\pi}{9} \cos\left(2\sin^{-1}\frac{1}{\sqrt{5}}\right)\right)$, $q = \sin^{-1}\left(\sin\frac{2\pi}{3}\right) + \cos^{-1}\left(\cos\left(\frac{7\pi}{6}\right)\right)$ then the quadratic equation whose roots are p, sec q is (Here inverse trigonometric functions take principal values)
- 1) $2\sqrt{3}x^2 - 6x + \sqrt{3} = 0$ 2) $\sqrt{3}x^2 - 4x + 2\sqrt{3} = 0$
 3) $\sqrt{3}x^2 - x - 2\sqrt{3} = 0$ 4) $x^2 - 4\sqrt{3}x + 4 = 0$
12. Find the value of $\lim_{x \rightarrow 0} \frac{e^x - x - 1}{x^2}$
- 1) $\frac{1}{2}$ 2) 1 3) 2 4) 0
13. If square matrix A is of order 3 then find the value of $|adj(adj A)|$, if $|A| = 2$
- 1) 2 2) 4 3) 8 4) 16





14. $f(x) = x^3 - x^2 f'(1) + x f''(2) - f'''(3), x \in R$ then which of the following is incorrect?
 1) $f'(1) = 3$ 2) $f'(3) = 10$ 3) $f(0) = -6$ 4) $f''(2) = 6$
15. **Statement - I:** $f: R \rightarrow R$ be a function such that $|f(x)| \leq x^2, \forall x \in R$ then $f(x)$ is differentiable at $x=0$.
Statement - II: $f: R \rightarrow R$ be a function such that $|f(x)| \leq x^p, \forall x \in R$ then $f(x)$ is differentiable at $x=0$ for any whole number 'p'
 1) Both statement - I and statement II are true
 2) Both statement- I and statement II are false
 3) Statement I is true and statement II is false
 4) Statement I is false and statement II is true
16. Let $y = f(x)$ satisfy the differential equation $\frac{dy}{dx} = 2xe^{-y}, \forall x \in R$. If $y'(1) = 1$ then the number of solutions of the equation $f(x) = f'(x)$ in $(0, \infty)$ is
 1) 1 2) 2 3) 3 4) zero
17. Match the following

	Column - I		Column - II
a)	The number of points of discontinuity of $f(x) = (\operatorname{sgn}(x))(\operatorname{sgn}(-x))$ in R is	i)	0
b)	The number of points of continuity of $f(x) = \begin{cases} e^{ x } - 1, x \text{ is rational} \\ \frac{1}{1+x^2}, x \text{ is irrational} \end{cases}$ in R is	ii)	1
c)	The number of points of discontinuity of $g(x) = \lim_{n \rightarrow \infty} \frac{(f(x))^n - 1}{(f(x))^n + 1}$ in $(0, \infty)$ where $f(x) = (x-1)(x-2)(x-3) + 1$ is	iii)	2
d)	The number of points of discontinuity of $f(x) = \max.([x], x)$ in R is, where $[.]$ represents the GIF	iv)	3

Which of the following match is correct?

- 1) $a - iii, b - iv, c - i, d - ii$ 2) $a - ii, b - iii, c - iv, d - i$
 3) $a - iii, b - i, c - iv, d - ii$ 4) $a - iv, b - i, c - ii, d - iii$





18. Let (α, β, γ) be the mirror image of $(1, -2, 3)$ in the line $\frac{x+1}{2} = \frac{y-3}{4} = \frac{z+2}{3}$ then
 $2\alpha + 4\beta + 3\gamma + 2 =$ _____
 1) 5 2) 11 3) 9 4) 10
19. How many out of the following statements are correct
STATEMENT I: a, b are non-zero real numbers then $\sqrt{ab} = \sqrt{a}\sqrt{b}$ if at least one of them is positive
STATEMENT II: a, b are non-zero real numbers then $\sqrt{ab} = \sqrt{a}\sqrt{b}$ if at least one of them is negative
STATEMENT III: a, b are non-zero real numbers then $\sqrt{\frac{a}{b}} = \frac{\sqrt{a}}{\sqrt{b}}$ if $ab > 0$ or $(a < 0, b > 0)$
 1) 0 2) 1 3) 2 4) 3
20. The number of distinct reflexive relations defined on set A, where set $A = \{1, 2, 3\}$, is
 1) 2^6 2) 2^2 3) 2^{12} 4) 2

SECTION-II**(NUMERICAL VALUE TYPE)**

This section contains **5 Numerical Value Type Questions**. The Answer should be within **0 to 9999**. If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i.e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11**).

Marking scheme: +4 for correct answer, 0 if not attempt and -1 in all other cases.

21. Find the maximum distance of the point $(-1, -1)$ from the points on circle
 $(x-5)^2 + (y-7)^2 = 9$
22. $\sin \beta = \frac{4 \sin^2 \alpha}{1 + 4 \sin^2 \alpha}, \sin \gamma = \frac{4 \sin^2 \beta}{1 + 4 \sin^2 \beta}, \sin \alpha = \frac{4 \sin^2 \gamma}{1 + 4 \sin^2 \gamma}$
 where $\alpha, \beta, \gamma \in R$, then the maximum value of $\frac{(\sin \alpha + \sin \beta + \sin \gamma)^2}{(\cos \alpha + \cos \beta + \cos \gamma)^2}$ is?
23. The number of solutions of the equation $\sin^2 x + (2 + 2x - x^2) \sin x - 3(x-1)^2 = 0$ in $\left[0, \frac{\pi}{2}\right]$ is α , and in $[-2\pi, 2\pi]$ is β then $\alpha + \beta =$ _____
24. If $a_1, a_2, a_3, \dots$ are in Geometric progression, $a_2 + a_4 + a_6 + \dots + a_{100} = 4$ and $a_1 + a_3 + a_5 + \dots + a_{99} = 2$ then find the value of constant ratio of the given Geometric progression
25. An online exam is attempted by 40 candidates out of which 15 are boys. The average marks obtained by boys is 10 with variance 2. The variance of marks obtained by 25 girls is also 2 and the average marks of all 40 candidates is 12.5. If μ is average marks of the 25 girls & σ^2 is variance of marks of all 40 candidates then $20\sigma^2 - 8\mu =$ _____





PHYSICS

Max Marks: 100

SECTION-I
(SINGLE CORRECT ANSWER TYPE)

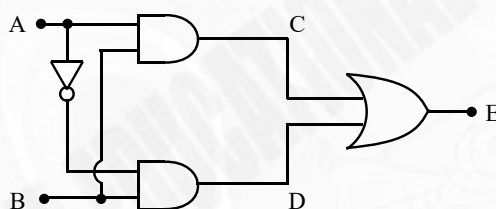
This section contains **20 Multiple Choice Questions**. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

26. The kinetic energies of photoelectrons emitted from a metal are K_1 and K_2 when it is irradiated with lights of wavelength λ_1 and λ_2 respectively. The work function of the metal is

1) $\frac{K_2\lambda_2 - K_1\lambda_1}{\lambda_1 - \lambda_2}$ 2) $\frac{K_1\lambda_1 - K_2\lambda_2}{\lambda_2 + \lambda_1}$ 3) $\frac{K_1\lambda_2 - K_2\lambda_1}{\lambda_2 - \lambda_1}$ 4) $\frac{K_1\lambda_2 + K_2\lambda_1}{\lambda_2 + \lambda_1}$

27. Truth table for the given circuit is



A	B	E	A	B	E	A	B	E	A	B	E
0	0	1	0	0	1	0	0	0	0	0	0
1) 0	1	0	2) 0	1	0	3) 0	1	1	4) 0	1	1
1	0	1	1	0	0	1	0	0	1	0	1
1	1	0	1	1	0	1	1	1	1	1	0

28. A physical quantity of the dimensions of length that can be formed out of c , G and $\frac{e^2}{4\pi\epsilon_0}$ is

[c is velocity of light, G is universal constant of gravitation and e is charge]

1) $\frac{1}{c^2} \left[G \frac{e^2}{4\pi\epsilon_0} \right]^{1/2}$ 2) $c^2 \left[G \frac{e^2}{4\pi\epsilon_0} \right]^{1/2}$ 3) $\frac{1}{c^2} \left[\frac{e^2}{G 4\pi\epsilon_0} \right]^{1/2}$ 4) $\frac{1}{c} G \frac{e^2}{4\pi\epsilon_0}$

29. In an ac circuit, the instantaneous current is zero, when the instantaneous voltage is maximum. In this case, the source may be connected to:

- a) Pure inductor b) Pure capacitor
c) Pure resistor d) Combination of an inductor and capacitor

Choose the correct answer from the options given below:

- 1) b, c and d only 2) a and b only 3) a, b, c only 4) a, b, d only

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Page 7



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JEE MAIN
2023

SINGARAJU
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ALL INDIA RANK 1

341

RANK

1

JEE Advanced
2023

VAJILALA
CHIVILAS REDDY

ALL INDIA RANK 1

341

RANK

1

NEET
2023

BORA VARUN
CHAKRAVARTI

ALL INDIA RANK 1

726

RANK

1

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30. The displacement of two identical particles executing SHM are represented by equations

$$x_1 = 4 \sin \left(10t + \frac{\pi}{6} \right) \text{ and } x_2 = 5 \cos \omega t$$

For what value of ω , the energy of both the particles become equal?

- 1) 16 unit 2) 6 unit 3) 4 unit 4) 8 unit

31. Given below are two statements

Statement (1): Mean free path of molecules of a gas decreases with increase in number density of the molecules

Statement (2): More the molecules per unit volume, more the chances of molecular collision. Therefore, means free path decreases.

- 1) Statement - (1) is false but statement (2) is true
 2) Statement - (1) and statement (2) are true
 3) Statement - (1) and statement (2) are false
 4) Statement - (1) is true but statement (2) is false

32. **Assertion:** Solids can support both longitudinal and transverse waves but only longitudinal waves can propagate in gases.

Reason: For the propagation of transverse waves, medium must also necessarily have the property of rigidity.

- 1) Both assertion and reason are true and the reason is the correct explanation of the assertion.
 2) Both assertion and reason are true but reason is not the correct explanation of the assertion.
 3) Assertion is true but reason is false.
 4) Assertion is false but reason is true.

33. **STATEMENT I:** Degree of freedom is number independent ways a gas molecule can store energy including translational, rotational and vibrational movements

STATEMENT II: Each quadratic term in total energy expression of a molecule is to be counted as a degree of freedom





- 1) Statement - (1) is false but statement (2) is true
- 2) Statement - (1) and statement (2) are true
- 3) Statement - (1) and statement (2) are false
- 4) Statement - (1) is true but statement (2) is false

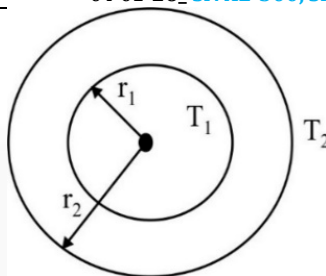
34. Match List - I with List - II

	List – I (Y vs X)		List – II (Shape of graph)
a)	Y= magnetic susceptibility X= absolute temperature(For paramagnetic materials)	I)	
b)	Y= magnetic field X= distance from the Axis of current carrying Cylindrical pipe	II)	
c)	Y- magnetic field X- Distance from The axis of current Carrying cylindrical Conductor	III)	
d)	Y- magnetic field X- Distance from the Centre of current carrying Circular ring along its axis	IV)	

- 1) (a) – II, (b) – III, (c) – IV, (d) – I
- 2) (a) – I, (b) – III, (c) – II, (d) – IV
- 3) (a) – II, (b) – III, (c) – I, (d) – II
- 4) (a) – III, (b) – II, (c) – IV, (d) – I

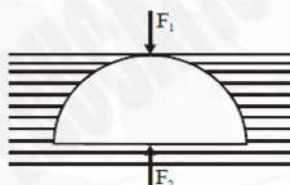
35. The figure shows a system of two concentric spheres of radii r_1 and r_2 kept at temperature T_1 and T_2 respectively. The radial rate of flow of heat in the substance between the concentric spheres is proportional to





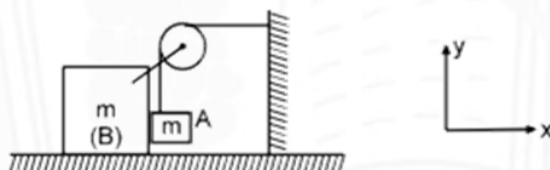
- 1) $\frac{r_1 r_2}{r_2 - r_1}$ 2) $(r_1 - r_2)^2$ 3) $\frac{r_1 - r_2}{r_1 r_2}$ 4) $\log\left(\frac{r_2}{r_1}\right)$

36. A solid hemisphere is just pressed below the liquid, the value of $\frac{F_1}{F_2}$ is (where F_1 and F_2 are the hydrostatic forces acting on the curved and flat surfaces of the hemisphere) (Neglect atmospheric pressure).



- 1) $\frac{1}{2}$ 2) $\frac{2}{3}$ 3) $\frac{1}{3}$ 4) $\frac{1}{4}$

37. In the system shown block A is released from rest. The acceleration of block A once released is

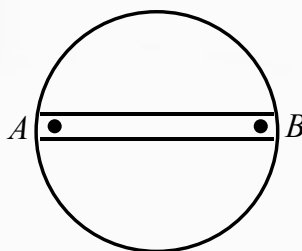


- 1) $\frac{g}{2}\hat{i} - \frac{g}{2}\hat{j}$ 2) $\frac{g}{3}\hat{i}$ 3) $\frac{g}{3}\hat{i} - \frac{g}{3}\hat{j}$ 4) $\frac{g}{2}\hat{i}$

38. A copper wire and a steel wire of the same diameter and length are connected end to end along the same line. A force is applied which stretches their combined length by 1 cm and the composite wire is at rest. Then the two wires have
- 1) The same stress and strain 2) The same stress but different strains
3) The same strain but different stresses 4) Different stresses and strains



39. Suppose a narrow, and straight tunnel passes through the center of earth as shown in the figure. Two small balls A and B are released simultaneously at the two ends of the tunnel. The balls collide with each other and after collision one ball comes to rest and the other ball just manages to escape the earth's gravitational field. If the coefficient of restitution between the balls is e , then select the correct option.



- 1) $e = \frac{\sqrt{3}}{2}$
- 2) $e = \frac{1}{\sqrt{2}}$
- 3) $\frac{1}{\sqrt{2}} < e < \frac{\sqrt{3}}{2}$
- 4) Information is insufficient to predict

40. Wheatstone bridge principle is used to measure the specific resistance (S_1) of given wire, having length L , radius r . If X is the resistance of wire, then specific resistance is;

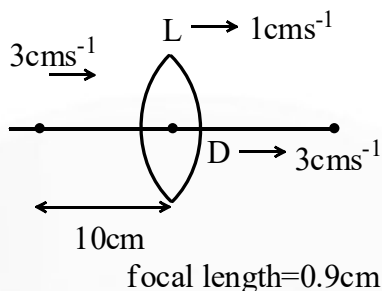
$S_i = X \left(\frac{\pi r^2}{L} \right)$. If the length of the wire gets double then the value of specific resistance will be:

- 1) $\frac{S_1}{4}$ 2) $2S_1$ 3) $\frac{S_1}{2}$ 4) S_1

- 41.** A body projected vertically upwards with a certain speed from top of a tower, reaches the ground in time t_1 . If it is projected vertically downwards with same speed from same point, it reach the ground in the time t_2 . Time required for it to reach the ground if it dropped from the same point and projected horizontally with same speed, respectively are

- 1) $\sqrt{t_1+t_2}, \sqrt{t_1-t_2}$ 2) $\sqrt{t_1+t_2}, \sqrt{t_1+t_2}$
3) $\sqrt{t_1 t_2}, \sqrt{t_1+t_2}$ 4) $\sqrt{t_1 t_2}, \sqrt{t_1 t_2}$

42. The figure shows the initial position of a point source of light s , a detector D and lens L . Now at $t=0$, all three starts moving towards right with different velocity as shown in figure. The times at which detector receives the maximum light, is (Assume that the detector is initially just touching the lens).

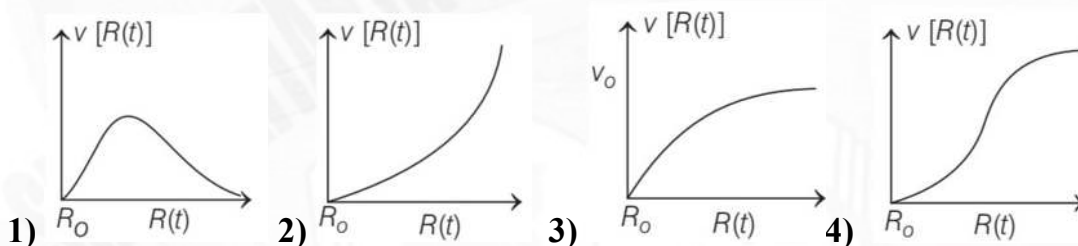


- 1) 5 s and 10 s 2) 0.5 s and 4.5 s 3) 2 s and 3 s 4) Only 10 s

43. In a thermodynamic process on an ideal diatomic gas, work done by the gas is η times the heat supplied ($\eta < 1$). The molar heat capacity of the gas for the process is

- 1) $\frac{5R}{2}$ 2) $5\eta \frac{R}{2}$ 3) $\frac{5R}{2(1-\eta)}$ 4) $\frac{5R(1-\eta)}{2}$

44. There is uniform spherically symmetric surface charge density at a distance R_0 from the origin. The charge distribution is initially at rest and starts expanding because of mutual repulsion. The figure that represents best the speed $v[R(t)]$ of the distribution as a function of its instantaneous radius $R(t)$ is



45. **ASSERTION:** No work is required to move a charge inside a charged conducting shell.

REASON: Electric field inside a uniformly charged solid sphere of radius R given by

$\frac{KQx}{R^3}$, where x is distance of point from center of sphere

- 1) If both assertion and reason are true and the reason is the correct explanation of the assertion.
- 2) If both assertion and reason are true but reason is not the correct explanation of the assertion.
- 3) If assertion is true but reason is false.
- 4) If assertion is false but reason is true.



JEE MAIN 2023

RANK 1

SINGARAJU

VENKAT KORDINIA

KODURU

KODURU

KODURU

RANK 1

JEE Advanced 2023

VAVILALA

CHIVILAS REDDY

KODURU

KODURU

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RANK 1

NEET 2023

BORA VARUN

CHAKRAVARTI

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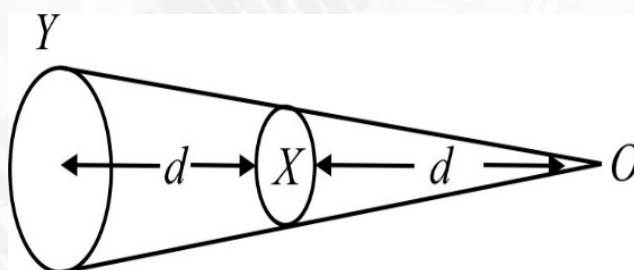
SECTION-II

(NUMERICAL VALUE TYPE)

This section contains **5 Numerical Value Type Questions**. The Answer should be within **0 to 9999**. If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i.e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11**).

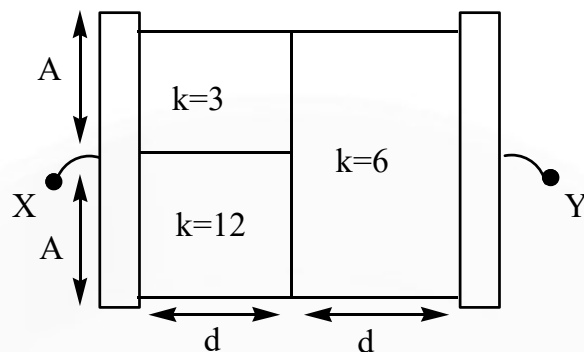
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46. A beam of ordinary light is incident on a system of 4 polaroids which are arranged in succession such that each polaroid is turned through 30° with respect to the preceding one. The percentage of the incident intensity that emerges out from the system is approximately
47. Two circular coils X and Y, having equal number of turns and carrying equal currents in the same sense subtend same solid angle at point O. If the smaller coil X is midway between O and Y and if we represent the magnetic induction due to bigger coil Y at O as B_y and that due to smaller coil X at O as B_x , then find the ratio $\frac{B_x}{B_y}$.

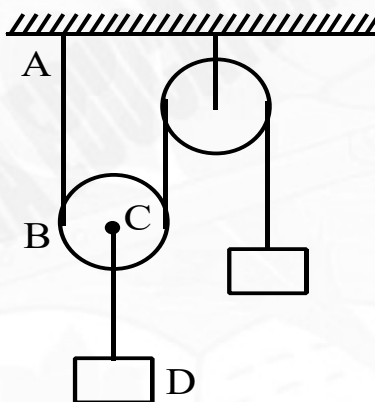


48. A body is executing simple harmonic motion. At a displacement 'x' from mean position its potential energy is $E_1 = 2J$ and at a displacement 'y' from mean position, its potential energy is $E_2 = 8J$. The potential energy 'E' (in J) at a displacement $(x + y)$ from mean position is.
49. Space between plates of parallel plate capacitor is filled with three dielectric slabs as shown. The different values of dielectric constants (k) are shown in the diagram for the three dielectrics. Equivalent dielectric constant for the arrangement shown (between points X and Y) is





50. Both the strings, show in figure, are made of same material and have same cross section. The pulleys are light. The wave speed of a transverse wave in the string AB is v_1 and in CD it is v_2 . Then the value of $10 \left(\frac{v_1}{v_2} \right)^2$ is



**CHEMISTRY****Max Marks: 100****SECTION-I
(SINGLE CORRECT ANSWER TYPE)**

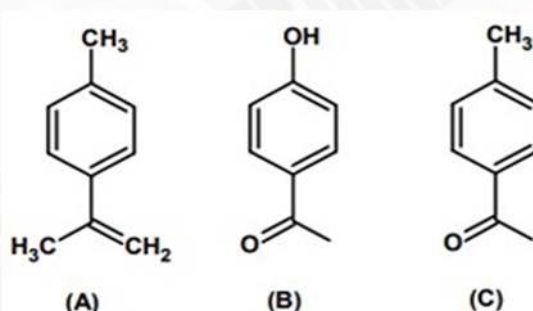
This section contains **20 Multiple Choice Questions**. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

51. If ' λ_0 ' is the threshold wavelength for photoelectric emission from a metal surface, ' λ ' is the wavelength of light falling on the surface of metal and ' m ' is the mass of electron, then the maximum speed of ejected electrons is given by

1) $\left[\frac{2h}{m} (\lambda_0 - \lambda) \right]^{\frac{1}{2}}$ 2) $\left[\frac{2hc}{m} (\lambda_0 - \lambda) \right]^{\frac{1}{2}}$ 3) $\left[\frac{2hc}{m} \left(\frac{\lambda_0 - \lambda}{\lambda \cdot \lambda_0} \right) \right]^{\frac{1}{2}}$ 4) $\left[\frac{2h}{m} \left(\frac{1}{\lambda_0} - \frac{1}{\lambda} \right) \right]^{\frac{1}{2}}$

52. A chromatography column packed with silica gel (used as stationary phase) is used for the separation of



The column was then eluted with a mixture of Dichloro methane-nitromethane in 80:20 ratio, the sequence of elution of A, B and C is (easily eluted compound first)

- 1) A, B, C 2) C, B, A 3) B, C, A 4) A, C, B
53. In which of the following arrangements the order is NOT according to the property indicated against it?
- 1) $Al^{3+} < Mg^{2+} < Na^+ < F^-$ - increasing ionic size
 2) $B < C < N < O$ - increasing first ionization enthalpy
 3) $I < Br < F < Cl$ - increasing order of electronic affinity
 4) $Li < Na < K < Rb$ - increasing metallic radius
54. Of the given anions, the strongest Bronsted base is

- 1) ClO^- 2) ClO_2^- 3) ClO_3^- 4) ClO_4^-

Sec: **Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT**

Page 15



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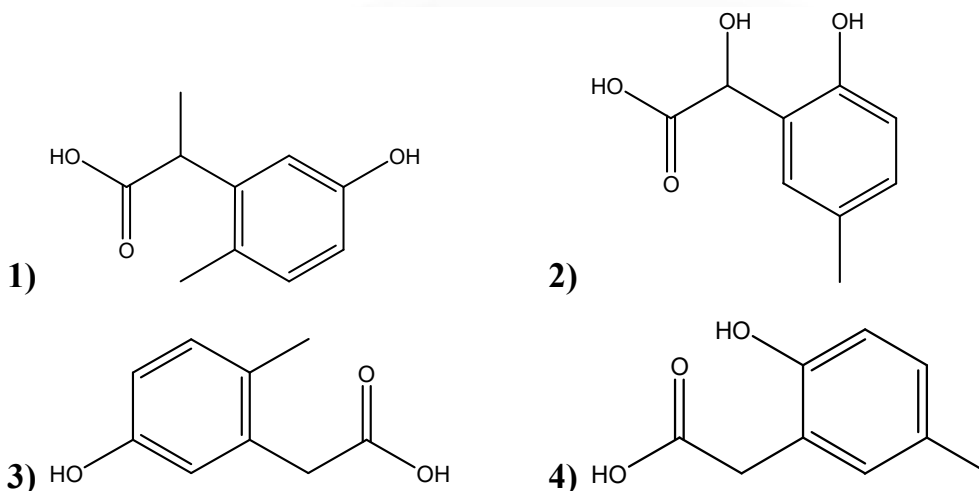
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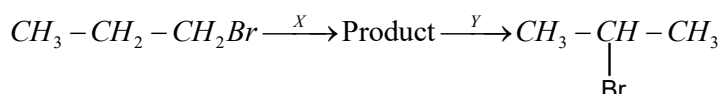
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55. P-Cresol reacts with chloroform in alkaline medium to give the compound 'A' which adds HCN to form the compound 'B'. The compound 'B' on acidic hydrolysis gives chiral carboxylic acid. The structure of carboxylic acid is



56. Identify the set of reagent/reaction conditions x and y in the following set of transformations.



- 1) X-dilute aqueous $NaOH$, $20^\circ C$; Y-HBr /acetic acid, $20^\circ C$
- 2) X- concentrated alcoholic $NaOH$, $80^\circ C$ Y-HBr /acetic acid, $20^\circ C$
- 3) X-dilute aqueous $NaOH$, $20^\circ C$; Y- $Br_2 / CHCl_3$, $0^\circ C$
- 4) X-concentrated alcoholic $NaOH$, $80^\circ C$; Y- $Br_2 / CHCl_3$, $0^\circ C$
57. Which of the following statement is not correct?
- 1) $B(OH)_3$ is acidic
- 2) Aqueous solution of Potash alum is acidic in nature
- 3) The decreasing order of Lewis acidic nature of BBr_3 , BCl_3 and BF_3 is $BBr_3 > BCl_3 > BF_3$
- 4) Al_2Cl_6 contains $Al - Al$ covalent bonds
58. The conductivity of a saturated aq.solution of $AgCl$ at 298 K is found to be $1.382 \times 10^{-2} \Omega^{-1} m^{-1}$ The ionic conductance of Ag^+ and Cl^- at infinite dilution are $61.92 \Omega^{-1} cm^2 mol^{-1}$ and $76.3 \Omega^{-1} cm^2 mol^{-1}$ respectively. The solubility of $AgCl$ is
- 1) $1 \times 10^{-1} mol L^{-1}$ 2) $1 \times 10^{-2} mol L^{-1}$ 3) $1 \times 10^{-3} mol L^{-1}$ 4) $1.9 \times 10^{-5} mol L^{-1}$
59. For the elementary reaction $M \rightarrow N$, the rate of disappearance of M increases by a factor of 8 upon doubling the concentration of M. The order of the reaction with respect to M is
- 1) 4 2) 3 3) 2 4) 1





60. Assertion (A): Aniline on nitration gives meta nitro aniline in maximum yield than other isomers.

Reason (R) : $\text{—}\overset{\oplus}{\text{N}}\text{H}_3$ acts as meta directing group.

- 1) Both A and R are true and R is the correct explanation of A
 - 2) Both A and R are true but R is not the correct explanation of A
 - 3) A is true but R is false
 - 4) A is false but R is true
61. The number of moles of KMnO_4 that will be needed to react completely with one mole of ferrous oxalate in acidic solution is
- 1) $\frac{3}{5}$
 - 2) $\frac{2}{5}$
 - 3) $\frac{4}{5}$
 - 4) 1
62. Arrange $\text{Ce}^{+3}, \text{La}^{+3}, \text{Pm}^{+3}, \text{Yb}^{+3}$ in increasing order of their ionic radii
- 1) $\text{Yb}^{+3} < \text{Pm}^{+3} < \text{Ce}^{+3} < \text{La}^{+3}$
 - 2) $\text{Ce}^{+3} < \text{Yb}^{+3} < \text{Pm}^{+3} < \text{La}^{+3}$
 - 3) $\text{Yb}^{+3} < \text{Pm}^{+3} < \text{La}^{+3} < \text{Ce}^{+3}$
 - 4) $\text{Pm}^{+3} < \text{La}^{+3} < \text{Ce}^{+3} < \text{Yb}^{+3}$
63. Benzene and toluene are nearly ideal solutions. If at a temperature $p_{\text{Benzene}}^0 = 120\text{mmHg}$ and $p_{\text{Toluene}}^0 = 50\text{mmHg}$. Then what will be the mole fraction of Toluene in vapour phase of this solution if mole fraction of Benzene in liquid phase of solution is 0.6.
- 1) 0.70
 - 2) 0.6
 - 3) 0.4
 - 4) 0.22
64. Assertion (A): $\text{K}_4[\text{Fe}(\text{CN})_6]$ is diamagnetic and $[\text{Fe}(\text{H}_2\text{O})_6]\text{Cl}_3$ is paramagnetic.
Reason (R): Hybridisation central metal in $\text{K}_4[\text{Fe}(\text{CN})_6]$ is sp^3d^2 , while in $[\text{Fe}(\text{H}_2\text{O})_6]\text{Cl}_3$ is d^2sp^3
Choose the correct option
- 1) Both A and R are Correct, and R is the correct explanation of A
 - 2) Both A and R are Correct, but R is not the correct explanation of A
 - 3) A is Correct but R is incorrect
 - 4) Both A and R are incorrect





65. **Statement-I** : All square planar complexes can exhibit geometrical isomerism.

Statement-II : In square planar complexes hybridization of central metal can be dsp^2 .

- 1) Both Statement 1 and Statement 2 are true, and Statement 2 is the correct explanation of Statement 1
- 2) Both Statement 1 and Statement 2 are true, but Statement 2 is NOT the correct explanation of Statement 1
- 3) Statement 1 is true but Statement 2 is false
- 4) Statement 1 is false but Statement 2 is true

66. **Statement-I**: Enthalpy and temperature are Intensive properties.

Statement-II: Intensive quantities does not depend on quantity of system.

- 1) Both Statement 1 and Statement 2 are true
- 2) Statement 1 is true but Statement 2 is false
- 3) Statement 1 is false but Statement 2 is true
- 4) Both Statement 1 and Statement 2 are false

67. Match the Column-I with Column-II

Species	Shape
(a) SO_4^{2-}	1) pyramidal
(b) ClO_3^-	2) linear
(c) C_3O_2	3) square planar
(d) $Ni(CO)_4$	4) tetrahedral

- 1) A-3,B-1,C-2,D-4
- 2) A-4,B-1,C-2,D-3
- 3) A-2,B-1,C-3,D-4
- 4) A-4,B-1,C-2,D-4

68. The Solubility of Calcium fluoride in water is $7.8 \times 10^{-4} g/L$ The value of $\log_{10} K_{sp}$ of calcium fluoride is

- 1) 4×10^{-15}
- 2) -14.4
- 3) 14.4
- 4) -4×10^{15}

69. Electromotive force of the following cell at 298K

$$Pt, H_2 / H^+ (0.001M) // H^+ (0.2M) / H_2, Pt \quad \left[\frac{2.303RT}{F} = 0.06 \right]$$

- 1) 120mv
- 2) 0.12 mv
- 3) 138mv
- 4) 0.138mv

70. Among the following the most basic compound is

- 1) benzylamine
- 2) aniline
- 3) acetanilide
- 4) p-nitroaniline





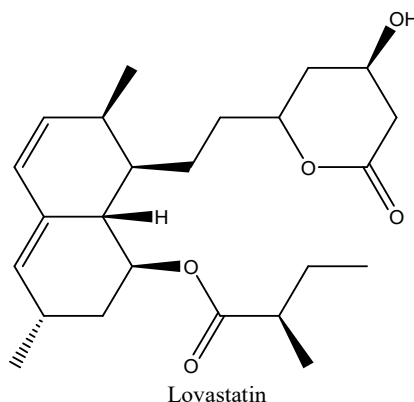
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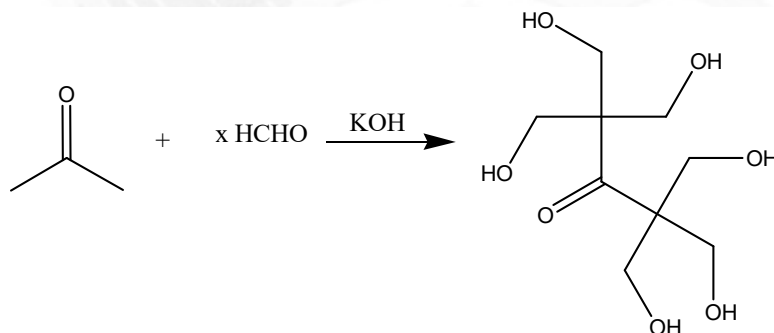
71. Lovastatin, a drug used to reduce the risk of cardio vascular diseases has the following structure



72. The number of stereogenic centers present in lovastatin is _____
How many of the following can show Tollens test

Formic acid, Acetophenone, Benzaldehyde, Acetone, Butanol, Ethanoic acid, Acetaldehyde

73.



x = moles of HCHO consumed. Value of (x) will be _____

74. The number of geometrical isomers possible for the complex $[Mabcd]$ is /are _____
(a,b,c,d are monodentate ligands)
75. A decapeptide (Molecular weight 796) on complete hydrolysis gives glycine (Molecular weight 75), alanine and phenylalanine. Glycine contributes 47.0% to the total weight of the hydrolyzed products. The number of glycine units present in the decapeptide is _____

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Page 19



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10

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